

# Multivariate Time Series (MTS) Forecasting

## The Core Challenge

MTS forecasting requires modelling two distinct but coupled types of dependencies simultaneously:

- **Intra-series (Temporal):** Dependencies within a single time series over time.
- **Inter-series (Spatial):** Correlations between different variables (e.g., traffic sensors, weather stations) at specific timestamps.

## The Problem

### How to unify these dependencies?

Real-world data exhibits unified spatiotemporal dynamics (e.g., traffic congestion propagation). Treating spatial and temporal modelling as separate steps violates these inherent connections and hinders performance.

# Existing Solutions and Why They Lack

## Common Approaches:

- **RNNs/CNNs (e.g., LSTM, TCN):** Focus heavily on temporal dependencies; struggle with complex spatial correlations among variables.
- **GNN-based Methods (e.g., STGCN, MTGNN):** Currently the State-of-the-Art. They stack:
  - Graph Networks (spatial)
  - Temporal Networks (temporal)

## Key Limitations:

- 1 **Decoupled Modeling:** Separate spatial/temporal modules create “uncertain compatibility” and high parameter complexity.
- 2 **High Complexity:** Many GNNs have  $O(N^2)$  complexity, making them expensive for large datasets.
- 3 **Static Priors:** Reliability often depends on pre-defined graph structures (which are often unknown or time-varying).

# Novel Solution: The Pure Graph Perspective

## Rethinking the Data Structure

Instead of separating dimensions, FourierGNN introduces the **Hypervariate Graph**.

- **Concept:** Treat every single data point  $x_t^{(n)}$  (variate  $n$  at time  $t$ ) as a unique node in one massive graph.
- **Structure:** A sliding window of  $N$  series over length  $T$  becomes a graph with  $N \times T$  nodes.
- **Connectivity:** Initialised as a fully-connected graph to capture all possible correlations.

*“This perspective considers spatiotemporal dynamics unitedly and reformulates forecasting into predictions on hypervariate graphs.”*

# Novel Solution: Fourier Graph Neural Network (FourierGNN)

To handle the large size of Hypervariate Graphs ( $N \times T$  nodes), the authors propose the **Fourier Graph Operator (FGO)**.

- 1 **Fourier Domain Operation:** Instead of expensive spatial convolution, the model operates in the frequency domain using DFT (Discrete Fourier Transform).

$$\text{Convolution Theorem: } \mathcal{F}(X * \kappa) = \mathcal{F}(X) \odot \mathcal{F}(\kappa)$$

- 2 **Efficiency:**
  - Spatial Graph Convolution:  $O(N^2)$  complexity.
  - **FourierGNN:** Reduces complexity to  $O(N \log N)$  (Log-linear) via FFT.
- 3 **Mechanism:** Stacking FGO layers in the Fourier space is mathematically equivalent to performing multi-order graph convolutions in the time domain, capturing long-range dependencies efficiently.

# Key Results and Performance

## Accuracy:

- Evaluated on 7 benchmarks (Traffic, Solar, Electricity, etc.).
- Achieved State-of-the-Art (SOTA) performance, outperforming Transformers (Informer) and GNNs (GraphWaveNet).
- **10% average accuracy improvement** reported in the paper.

## Efficiency:

- **Fewer Parameters:**  $\approx 20\%$  fewer parameters than lightweight baselines.
- **Faster Training:** Significant reduction in training time due to  $O(N \log N)$  complexity.

## Reproduction Analysis

Qualitative analysis confirms the model's ability to capture periodic patterns, though performance is sensitive to data preprocessing (scaling/normalisation).

## Limitations of FourierGNN

- **Loss of Sequential Information:** Treating every time-step as a node makes the graph *order-insensitive*. Permuting the time sequence results in an identical graph structure, losing temporal context.
- **Attention Dilution:** The **Fully-Connected** structure treats all connections uniformly. This “over-smooths” the data, washing out distinct local patterns with global noise.

## What we think?

**Proposal:** Replace Hypervariate Graphs with **Hyperspectral Graphs**.

- **The Method:** Embeds sequential order into frequency signals *before* graph construction.
- **Intuitive Benefit:**
  - *Preserves Time:* Embedding frequency locks the sequential order.
  - *Sharper Focus:* Allow the model to capture the global trends without blurring out specific, local features.

# Contributions

**Ayush Rawat** - Report, Dataset Sourcing, Code

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